



CS464 - Data and Network Security

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BLOCK CIPHER OPERATION

This chapter continues our discussion of symmetric ciphers. We begin with the topic of multiple encryption, looking in particular at the most widely used multiple-encryption scheme: triple DES.



The chapter next turns to the subject of block cipher modes of operation. We find that there are a number of different ways to apply a block cipher to plaintext, each with its own advantages and particular applications.

LEARNING OBJECTIVES

1

Analyze the security of multiple encryption schemes.

2

Explain the meet-in-the-middle attack.

3

Compare and contrast ECB, CBC, CFB, OFB, and counter modes of operation.

4

Present an overview of the XTS-AES mode of operation.



After studying this chapter, you should be able to:

7.1 MULTIPLE ENCRYPTION AND TRIPLE DES



Because of its vulnerability to brute-force attack, DES, once the most widely used symmetric cipher, has been largely replaced by stronger encryption schemes.

1

New Algorithm Approach

Design a completely new algorithm that is resistant to both cryptanalytic and brute-force attacks, of which AES is a prime example.

2

Multiple Encryption Approach

Use multiple encryption with DES and multiple keys, which preserves the existing investment in software and equipment.

DOUBLE DES



Encryption

Ciphertext Formula:

$$C = E(K_2, E(K_1, P))$$

- ✓ Two encryption stages
- ✓ Two different keys: K_1 and K_2
- ✓ Inner encryption with K_1 , outer with K_2



Decryption

Plaintext Formula:

$$P = D(K_1, D(K_2, C))$$

- ✓ Keys applied in reverse order
- ✓ Outer decryption with K_2 , inner with K_1
- ✓ Reverses the encryption process

Note: This scheme appears to involve a key length of $56 \times 2 = 112$ bits, resulting in a dramatic increase in cryptographic strength.

REDUCTION TO A SINGLE STAGE



Hypothetical Vulnerability

Suppose it were true for DES, for all 56-bit key values, that given any two keys K_1 and K_2 , it would be possible to find a key K_3 such that:

$$E(K_2, E(K_1, P)) = E(K_3, P)$$



Consequence

If this were true, then double encryption, and indeed any number of stages of multiple encryption with DES, would be useless.



Impact

The result would be equivalent to a single encryption with a single 56-bit key.

Note: On the face of it, it does not appear that Equation (7.1) is likely to hold, which was not proven until 1992.

PERMUTATION MAPPING



DES as a Permutation

- ✓ Encryption with DES is a mapping of 64-bit blocks to 64-bit blocks
- ✓ The mapping can be viewed as a permutation
- ✓ DES encryption with a specific key maps each of the 2^{64} possible input blocks into a unique 64-bit block



Uniqueness Requirement

If two given input blocks mapped to the same output block, decryption to recover the original plaintext would be impossible.

Key Insight: The permutation property ensures that every input block has a unique output block for a given key.

NUMBER OF POSSIBLE MAPPINGS



All Possible Permutations

With 2^{64} possible inputs, the number of different mappings is:

$$(2^{64})!$$

Numerical value: $10^{(3.4738 \times 10^{20})}$



DES Mappings

DES defines one mapping for each key:

$$2^{56}$$

Approximate value: 10^{17}



Conclusion

It is reasonable to assume that if DES is used twice with different keys, it will produce one of the many mappings that are not defined by a single application of DES.

MEET-IN-THE-MIDDLE ATTACK



Double DES Vulnerability

The use of double DES results in a mapping that is not equivalent to a single DES encryption. However, there is a way to attack this scheme.



Origin

The algorithm, known as a meet-in-the-middle attack, was first described in [DIFF77].



Universal Attack

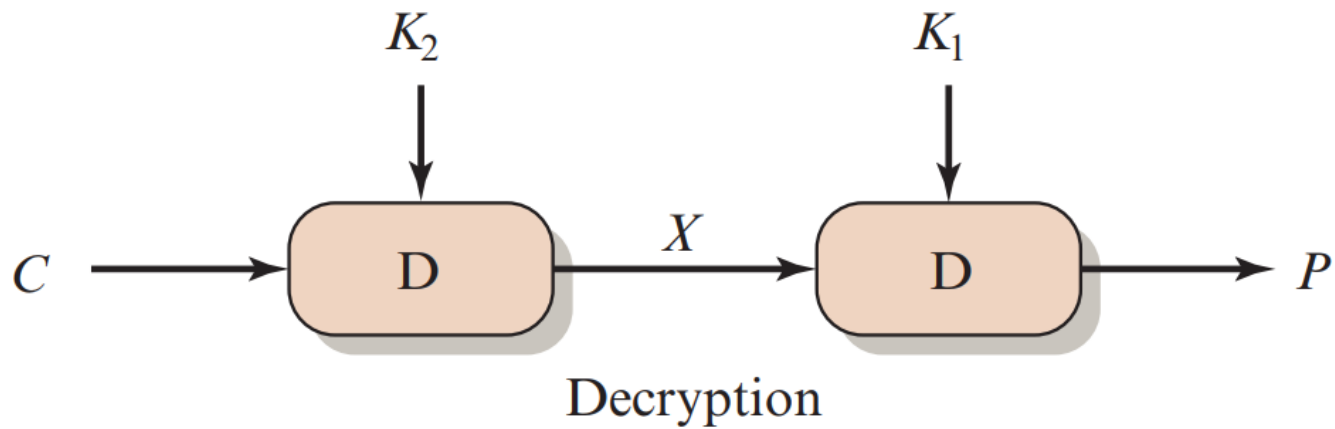
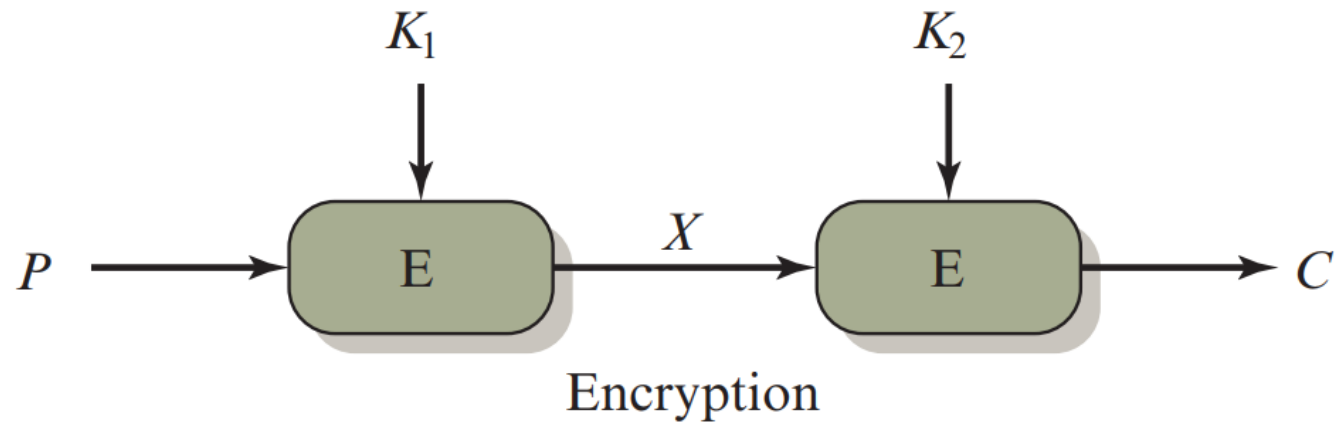
The attack does not depend on any particular property of DES but will work against any block encryption cipher.



Attack Principle

The attack is based on the observation that if we have $C = E(K_2, E(K_1, P))$, then $X = E(K_1, P) = D(K_2, C)$.

FIGURE 7.1A - MULTIPLE ENCRYPTION



(a) Double encryption

MEET-IN-THE-MIDDLE ATTACK - STEP 1

1

Encryption Phase

- Given a known plaintext-ciphertext pair (P, C)
- Encrypt P for all 2^{56} possible values of K_1
- Compute $X = E(K_1, P)$ for each K_1
- Store these results in a table
- Sort the table by the values of X



Computation

For each K_1 from 0 to $2^{56} - 1$:

$$X = E(K_1, P)$$

Total: 2^{56} encryption operations



Storage Requirement

Table stores 2^{56} entries of the form (K_1, X)

MEET-IN-THE-MIDDLE ATTACK - STEP 2

2

Decryption & Matching Phase

- Decrypt C using all 2^{56} possible values of K_2
- Compute $X' = D(K_2, C)$ for each K_2
- Check each X' against the sorted table from Step 1
- If a match occurs, candidate keys (K_1, K_2) are found
- Test the candidate keys against a new known plaintext-ciphertext pair
- If the keys produce the correct ciphertext, accept them as the correct keys



Matching Process

For each K_2 from 0 to $2^{56} - 1$:

$$X' = D(K_2, C)$$

Search table for X' value



Validation

Candidate keys must be verified using additional known plaintext-ciphertext pairs to eliminate false alarms.

MEET-IN-THE-MIDDLE ATTACK - ANALYSIS



Cryptographic Space Analysis

- **Double DES Key Space**

2^{112} possible keys ($56 \text{ bits} \times 2$)

- **Ciphertext Space**

2^{64} possible ciphertext values for any given plaintext P

- **Average Keys per Ciphertext**

Maximum number of different 112-bit keys that could produce a given ciphertext C



Calculation

Keys per Ciphertext =

$$2^{112} / 2^{64}$$

=

$$2^{48}$$



Implication

The attack can produce approximately 2^{48} false alarms on the first known plaintext-ciphertext pair.

FALSE ALARM RATE



False Alarm Analysis

- **First Known Pair (P, C)**

Produces approximately 2^{48} false alarms

- **With Additional 64 Bits**

False alarm rate reduced to $2^{48} - 64 = 2^{16}$

- **Two Known Pairs**

Probability of determining correct keys = $1 - 2^{-16}$



Attack Complexity

Known plaintext attack succeeds against double DES (112-bit key) with effort on the order of:

2^{56}

Not much more than 2^{55} required for single DES



Key Insight

The meet-in-the-middle attack effectively reduces the security of double DES to approximately the same level as single DES.

TRIPLE DES WITH TWO KEYS



Defense Against Meet-in-the-Middle Attack

- Use three stages of encryption with three different keys
- Using DES as the underlying algorithm
- Commonly referred to as 3DES or Triple Data Encryption Algorithm (TDEA)



Two Versions of 3DES

1

Two-key version

2

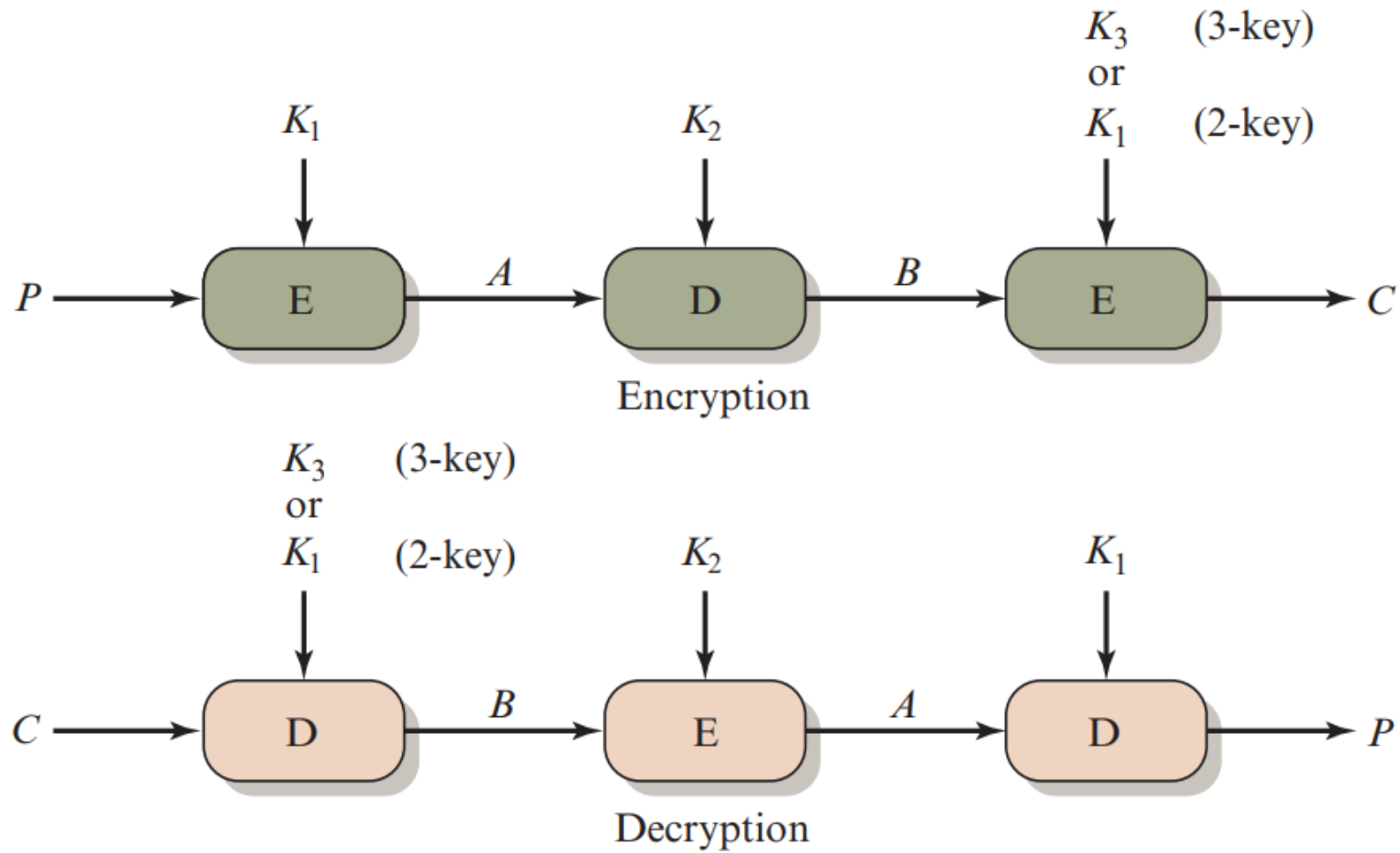
Three-key version



NIST Standard

NIST SP 800-67 defines the two-key and three-key versions. We examine the two-key version first, then the three-key version.

FIGURE 7.1B - TRIPLE ENCRYPTION



(b) Triple encryption

TWO-KEY TRIPLE ENCRYPTION DEFINITION

Origin

Two-key triple encryption was first proposed by Tuchman.

Encrypt-Decrypt-Encrypt (EDE) Sequence

Encryption:

$$C = E(K_1, D(K_2, E(K_1, P)))$$

Decryption:

$$P = D(K_1, E(K_2, D(K_1, C)))$$

Design Choice

There is no cryptographic significance to the use of decryption for the second stage.

Backward Compatibility

The only advantage is that it allows users of 3DES to decrypt data encrypted by users of the older single DES.

Compatibility Proof

$$C = E(K_1, D(K_1, E(K_1, P))) = E(K_1, P)$$

$$P = D(K_1, E(K_1, D(K_1, C))) = D(K_1, C)$$

BACKWARD COMPATIBILITY EXPLANATION



Purpose of EDE

The EDE sequence allows users of 3DES to decrypt data encrypted by users of the older single DES.



Key Condition

If we set $K_2 = K_1$, then the decryption stage simplifies.



Encryption Compatibility

With $K_2 = K_1$:

$$\begin{aligned} C &= E(K_1, D(K_1, E(K_1, P))) \\ &= E(K_1, P) \end{aligned}$$



Decryption Compatibility

With $K_2 = K_1$:

$$\begin{aligned} P &= D(K_1, E(K_1, D(K_1, C))) \\ &= D(K_1, C) \end{aligned}$$

SECURITY ANALYSIS OF 3DES



No Practical Attacks

Currently, there are no practical cryptanalytic attacks on 3DES.



Standard Adoption

3DES with two keys is a relatively popular alternative to DES and has been adopted for use in key management standards:

- ANSI X9.17
- ISO 8732.1



Brute-Force Key Search

Coppersmith notes that the cost of a brute-force key search on 3DES is on the order of:

$$2^{112} \approx (5 \times 10^{33})$$



Differential Cryptanalysis

The cost of differential cryptanalysis suffers an exponential growth compared to single DES:

$$> 10^{52}$$

BRUTE-FORCE KEY SEARCH COST



3DES Key Search

Coppersmith notes that the cost of a brute-force key search on 3DES is on the order of:

$$2^{112}$$

$$\approx (5 \times 10^{33})$$



Significance

This represents a significant increase in computational effort compared to single DES.



Single DES Comparison

Single DES requires approximately:

$$2^{55}$$

operations



Exponential Growth

The exponential growth in difficulty makes 3DES much more resistant to brute-force attacks than single DES.

MERKLE AND HELLMAN ATTACK PROPOSAL

Attack Significance

It is worth looking at several proposed attacks on 3DES that, although not practical, give a flavor for the types of attacks that have been considered and that could form the basis for more successful future attacks.

First Serious Proposal

The first serious proposal came from Merkle and Hellman.



Attack Strategy

Their plan involves:

- Finding plaintext values that produce a first intermediate value of **A = 0**
- Using the meet-in-the-middle attack to determine the two keys



Effort and Requirements

Level of effort:

2^{56}

Requires:

2^{56} chosen pairs

This is a number unlikely to be provided by the holder of the keys.

KNOWN-PLAINTEXT ATTACK OVERVIEW



Attack Type

A known-plaintext attack is outlined. This method is an improvement over the chosen-plaintext approach but requires more effort.



Key Observation

The attack is based on the observation that if we know **A** and **C** (refer to Figure 7.1b), then the problem reduces to that of an attack on double DES.



Attacker's Challenge

Of course, the attacker does not know **A**, even if **P** and **C** are known, as long as the two keys are unknown.



Attack Strategy

However, the attacker can choose a potential value of **A** and then try to find a known (**P**, **C**) pair that produces **A**.



This approach allows the attacker to test candidate keys without needing the chosen plaintext-ciphertext pairs required by Merkle and Hellman's method.

TRIPLE DES WITH THREE KEYS



Researcher Recommendation

Although the attacks just described appear impractical, anyone using two-key 3DES may feel some concern. Thus, many researchers now feel that three-key 3DES is the preferred alternative.



Definition

Three-key 3DES is defined as:

$$C = E(K_3, D(K_2, E(K_1, P)))$$



NIST Recommendation

In SP 800-57, Part 1, NIST recommends that 2-key 3DES be retired as soon as practical and replaced with 3-key 3DES.



Enhanced Security

The three-key version provides significantly stronger security by using three independent keys for the EDE sequence.

BACKWARD COMPATIBILITY WITH THREE KEYS



Compatibility Mechanism

Backward compatibility with DES is provided by putting $K_3 = K_2$ or $K_1 = K_2$.



Functionality

By setting certain keys equal to each other, the three-key 3DES can be reduced to function like single DES or two-key 3DES.



Key Configurations

- **$K_3 = K_2$:**
Reduces to two-key 3DES
- **$K_1 = K_2 = K_3$:**
Reduces to single DES



Migration Benefits

This feature allows systems that implement three-key 3DES to remain compatible with older systems that use single DES or two-key 3DES, facilitating a smooth transition during migration.

SECURITY STRENGTH ANALYSIS



Expected Strength

One might expect that 3TDEA would provide $56 \times 3 = 168$ bits of strength.



Merkle-Hellman Attack

This attack, proposed by Merkle and Hellman, demonstrates that even three-key 3DES does not provide the full 168-bit security that might be expected from three independent 56-bit keys.



Actual Strength

However, there is an attack on 3TDEA that reduces the strength to the work that would be involved in exhausting a 112-bit key.



Current Assessment

Nevertheless, 112-bit security remains very strong and is considered secure for most applications.

APPLICATIONS OF THREE-KEY 3DES



Internet Applications

A number of Internet-based applications have adopted three-key 3DES, including PGP and S/MIME, both discussed in Chapter 21.



Data Protection

These applications use the strong security provided by three-key 3DES to protect sensitive data in various contexts, such as email encryption and secure communication.



PGP & S/MIME

- Pretty Good Privacy (PGP) for email encryption
- Secure/Multipurpose Internet Mail Extensions (S/MIME)
- Both provide secure email communication



Community Acceptance

The adoption of three-key 3DES in widely used applications demonstrates its acceptance as a secure encryption standard in the cryptographic community.

BLOCK CIPHER MODES OF OPERATION



Multiple Applications

The chapter next turns to the subject of block cipher modes of operation. We find that there are a number of different ways to apply a block cipher to plaintext, each with its own advantages and particular applications.



Handling Large Plaintext

If the amount of plaintext to be encrypted is greater than b bits, then the block cipher can still be used by breaking the plaintext up into b -bit blocks.



Block Cipher Definition

A block cipher takes a fixed-length block of text of length b bits and a key as input and produces a b -bit block of ciphertext.



NIST Standards

NIST (SP 800-38A) has defined five modes of operation that are intended for use with any symmetric block cipher, including triple DES and AES.

ELECTRONIC CODEBOOK (ECB) MODE



Simplest Mode

The simplest mode is the electronic codebook (ECB) mode, in which plaintext is handled one block at a time and each block of plaintext is encrypted using the same key.



Imaginary Codebook

Therefore, we can imagine a gigantic codebook in which there is an entry for every possible b-bit plaintext pattern showing its corresponding ciphertext.



Codebook Concept

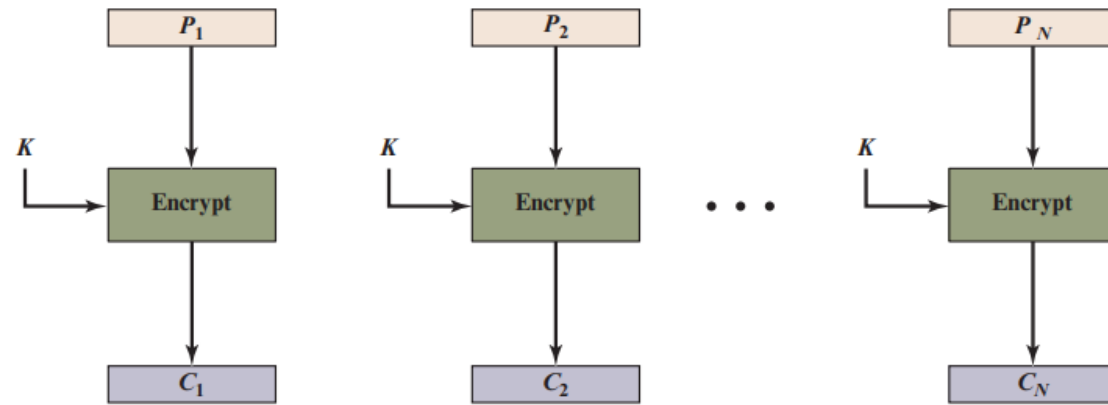
The term codebook is used because, for a given key, there is a unique ciphertext for every b-bit block of plaintext.



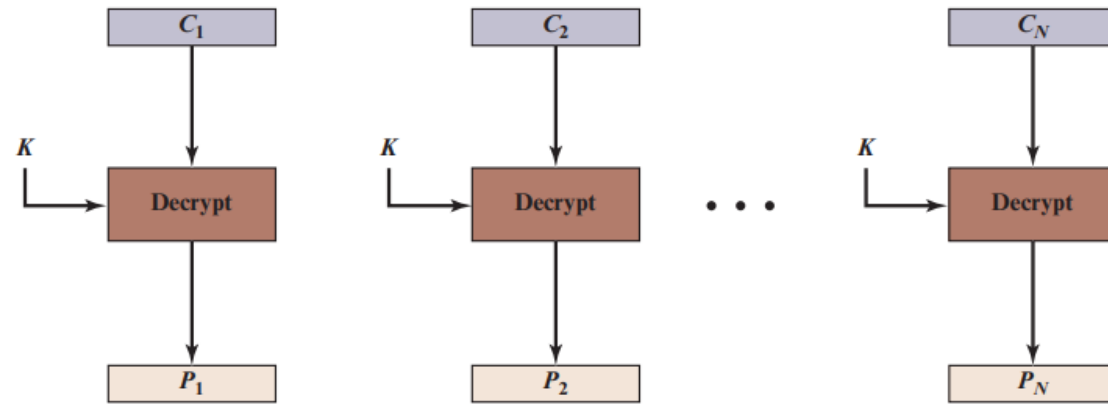
Block Processing

- Plaintext handled one block at a time
- Each block encrypted with same key
- Unique ciphertext per plaintext block

FIGURE 7.3 - ELECTRONIC CODEBOOK (ECB) MODE



(a) Encryption



(b) Decryption

Figure 7.3 Electronic Codebook (ECB) Mode

ECB MODE ENCRYPTION AND DECRYPTION



Message Handling

For a message longer than b bits, the procedure is simply to break the message into b -bit blocks, padding the last block if necessary.



Encryption Process

Plaintext blocks:

$P_1, P_2, \dots, P_n$

Encryption formula:

$$C_j = E(K, P_j)$$

for $j = 1, \dots, N$



Decryption Process

Decryption is performed one block at a time, always using the same key.



Decryption Formula

Ciphertext blocks:

$C_1, C_2, \dots, C_n$

Decryption formula:

$$P_j = D(K, C_j)$$

for $j = 1, \dots, N$

ECB MODE SECURITY CHARACTERISTICS



Critical Vulnerability

The most significant characteristic of ECB is that if the same b-bit block of plaintext appears more than once in the message, it always produces the same ciphertext.



Pattern Preservation

This property means that patterns in the plaintext are preserved in the ciphertext, which can leak information about the structure of the plaintext.



Repetitive Elements

If the message has repetitive elements with a period of repetition a multiple of b bits, then these elements can be identified by the analyst.



Analysis Impact

- Cryptanalyst can identify repetitive patterns
- Plaintext structure revealed through ciphertext
- May allow substitution or rearrangement of blocks

ECB MODE LIMITATIONS



Limited Applicability

The ECB mode should be used only to secure messages shorter than a single block of underlying cipher (64 bits for 3DES and 128 bits for AES), such as to encrypt a secret key.



Lengthy Messages Risk

For lengthy messages, the ECB mode may not be secure.



Minimal Practical Value

Because in most cases messages are longer than the encryption block mode, this mode has a minimum practical value.



Structured Messages Vulnerability

If the message is highly structured, it may be possible for a cryptanalyst to exploit these regularities.

- Known plaintext-ciphertext pairs can be obtained
- Repetitive patterns can be identified

CRITERIA FOR EVALUATING MODES OF OPERATION



Overhead

The additional operations for the encryption and decryption operation when compared to encrypting and decrypting in the ECB mode.



Error Propagation

The property that an error in the i -th ciphertext block is inherited by the i -th and all subsequent plaintext blocks.



Error Recovery

The property that an error in the i -th ciphertext block is inherited by only a few plaintext blocks after which the mode resynchronizes.



Diffusion

How the plaintext statistics are reflected in the ciphertext. Low entropy plaintext blocks should not be reflected in the ciphertext blocks.



These criteria are listed in [KNUD00] for evaluating and constructing block cipher modes of operation that are superior to ECB.

CIPHER BLOCK CHAINING (CBC) MODE



Overcoming ECB Deficiencies

To overcome the security deficiencies of ECB, we would like a technique in which the same plaintext block, if repeated, produces different ciphertext blocks.



Chaining Concept

A simple way to satisfy this requirement is the cipher block chaining (CBC) mode.



Encryption Input

In this scheme, the input to the encryption algorithm is the XOR of the current plaintext block and the preceding ciphertext block.

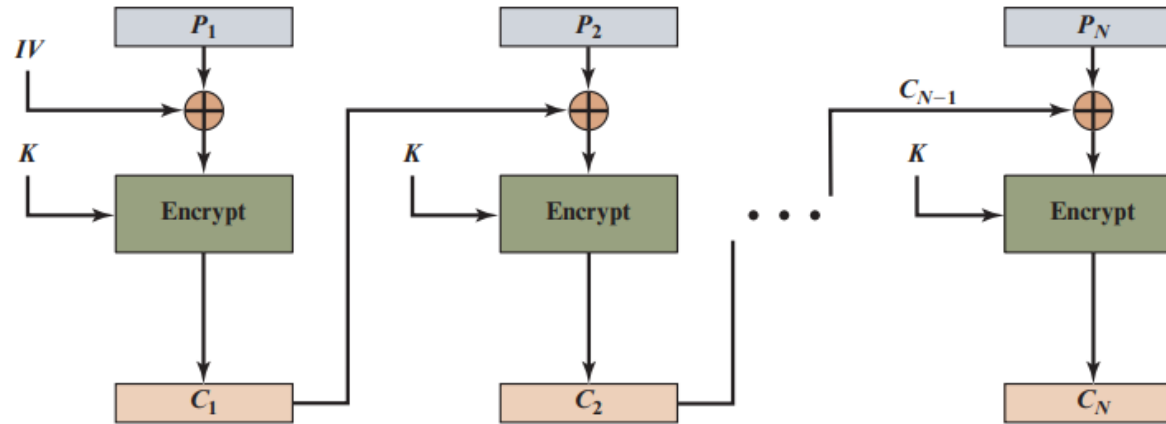
Same key used for each block



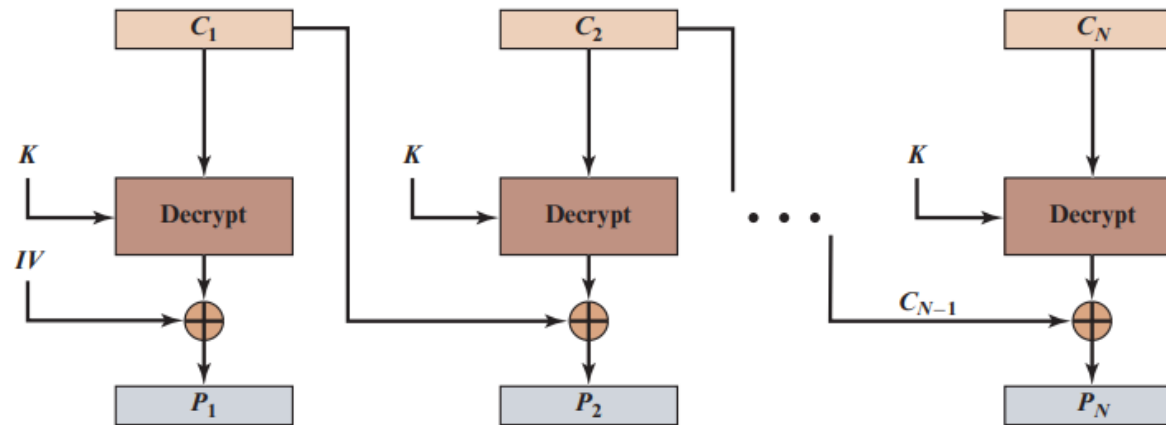
Chained Processing

In effect, we have chained together the processing of the sequence of plaintext blocks.

FIGURE 7.4 - CIPHER BLOCK CHAINING (CBC) MODE



(a) Encryption



(b) Decryption

CBC MODE ENCRYPTION PROCESS



Randomized Input

The input to the encryption function for each plaintext block bears no fixed relationship to the plaintext block.



Pattern Protection

Therefore, repeating patterns of b bits are not exposed.



Padding Requirement

As with the ECB mode, the CBC mode requires that the last block be padded to a full b bits if it is a partial block.

Encryption Algorithm

The input to the encryption algorithm is the XOR of the current plaintext block and the preceding ciphertext block.

Same key used for each block



Key Advantage

In effect, we have chained together the processing of the sequence of plaintext blocks, making each ciphertext block dependent on all preceding plaintext blocks.

CBC MODE DECRYPTION PROCESS



Decryption Steps

For decryption, each cipher block is passed through the decryption algorithm. The result is XORed with the preceding ciphertext block to produce the plaintext block.



Reversal Process

The decryption process reverses the encryption steps exactly. The XOR operation with the preceding ciphertext block ensures that each plaintext block is correctly recovered.



Mathematical Proof

Encryption:

$$C_j = E(K, [C_{j-1} \oplus P_j])$$

Decryption:

$$D(K, C_j) = C_{j-1} \oplus P_j$$

Final Result:

$$C_{j-1} \oplus D(K, C_j) = P_j$$



Key Property

$D(K, E(K, X)) = X$, which ensures that the decryption algorithm perfectly reverses the encryption algorithm's output.

Chaining Effect

Each plaintext block depends on the preceding ciphertext block, creating a chain that makes the CBC mode more secure than ECB.

CBC MODE MATHEMATICAL DEFINITION



Encryption

First Block:

$$C_1 = E(K, [P_1 \oplus IV])$$

Subsequent Blocks ($j = 2, \dots, N$):

$$C_j = E(K, [P_j \oplus C_{j-1}])$$



Decryption

First Block:

$$P_1 = D(K, C_1) \oplus IV$$

Subsequent Blocks ($j = 2, \dots, N$):

$$P_j = D(K, C_j) \oplus C_{j-1}$$



Key Points

- IV is XORed with first plaintext block
- Each ciphertext block depends on previous ciphertext block
- Same key K used for all blocks



Initialization Vector (IV)

IV is a data block that is the same size as the cipher block. It is XORed with the first plaintext block before encryption.

INITIALIZATION VECTOR (IV) REQUIREMENTS



Shared Knowledge

The IV must be known to both the sender and receiver but be unpredictable by a third party.



Maximum Security

For maximum security, the IV should be protected against unauthorized changes.



Unpredictability

For any given plaintext, it must not be possible to predict the IV that will be associated to the plaintext in advance of the generation of the IV.



IV Protection Method

This could be done by sending the IV using ECB encryption.

IV SECURITY CONSIDERATIONS



IV Manipulation Attack

If an opponent is able to fool the receiver into using a different value for IV, then the opponent is able to invert selected bits in the first block of plaintext.



Critical Vulnerability

Predictable changes to IV result in predictable changes to the recovered plaintext P1, allowing attackers to manipulate the first block of decrypted data.



Bit Manipulation Analysis

Encryption:

$$C_1 = E(K, [IV \oplus P_1])$$

Decryption:

$$P_1 = IV \oplus D(K, C_1)$$

Bit-level operation (i-th bit):

$$P_1[i] = IV[i] \oplus D(K, C_1)[i]$$

With bit complementation (denoted by '):

$$P_1[i]' = IV[i]' \oplus D(K, C_1)[i]$$

IV GENERATION METHODS



Unpredictability is Key

So long as it is unpredictable, the specific choice of IV is unimportant.



NIST Recommendation

SP 800-38A recommends two possible methods for IV generation.



Method 1: Encrypt a Nonce

Apply the encryption function, under the same key that is used for the encryption of the plaintext, to a nonce.



Nonce Definition

A data block that is unique to each execution of the encryption operation. It must have at most a negligible chance of repeating.



Nonce Examples

- Random value generated anew
- Sequence number
- Timestamp
- Combination of these

BLOCK CIPHER MODES OF OPERATION

Table 7.1 Block Cipher Modes of Operation

Mode	Description	Typical Application
Electronic Codebook (ECB)	Each block of plaintext bits is encoded independently using the same key.	• Secure transmission of single values (e.g., an encryption key)
Cipher Block Chaining (CBC)	The input to the encryption algorithm is the XOR of the next block of plaintext and the preceding block of ciphertext.	• General-purpose block-oriented transmission • Authentication
Cipher Feedback (CFB)	Input is processed s bits at a time. Preceding ciphertext is used as input to the encryption algorithm to produce pseudorandom output, which is XORed with plaintext to produce next unit of ciphertext.	• General-purpose stream-oriented transmission • Authentication
Output Feedback (OFB)	Similar to CFB, except that the input to the encryption algorithm is the preceding encryption output, and full blocks are used.	• Stream-oriented transmission over noisy channel (e.g., satellite communication)
Counter (CTR)	Each block of plaintext is XORed with an encrypted counter. The counter is incremented for each subsequent block.	• General-purpose block-oriented transmission • Useful for high-speed requirements



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